Addition & Subtraction

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Addition & subtraction

How do I add large numbers without a calculator?

- There are a variety of written methods that can be used to add large numbers
  - The order in which numbers are added is **not** important
- The **column method** is the most commonly used hand written method
  - The numbers are written one number above the other,
    - Line up the digits using **place value columns**
  - Add each pair of corresponding digits from the top and bottom rows (work **right to left**)
  - If the result is a single digit
    - write the result in the relevant place value column below the line
  - If the result is a 2 digit number
    - the ones are written in the relevant place value column below the line
    - the tens are carried to the top of the next column
  - If the addition of the final pair of digits results in a 2 digit number
    - write both digits below the line
- For example, the addition 9789 + 563 = 10 352

$$\begin{array}{r} {\scriptstyle 1\,1\,1} \\ 9\,7\,8\,9 \\ +\quad 5\,6\,3 \\ \hline 1\,0\,3\,5\,2 \end{array}$$

How do I subtract large numbers?

- A variety of written methods exist, but you only need to know one
  - The order in which two numbers are subtracted **is** important so ensure the calculation is the right way round
- The **column method** is the most commonly used hand written method
  - The numbers are written one number above the other
    - Line up the digits using **place value columns**

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

corresponding digit in the top value (work **right to left**)
    - If the digit being subtracted is bigger than the one it is subtracted from
        - **"borrow ten"** from the next column to the left
- For example, 392 - 28 = 364

$$\begin{array}{r} 3\overset{8}{\cancel{9}}\overset{1}{2} \\ -\ \ 28 \\ \hline 364 \end{array}$$

- Another method is **counting up**
    - E.g. to find 673 - 289 start at 289 and count up
        - Add 1 to get 290
        - Add 10 to get 300
        - Add 300 to get 600
        - Add 73 to get to 673
    - then add the steps together
        - 1+10+300+73=384
        - 673−289=384

What words are used for addition and subtraction?

- **Addition** may be phrased using the words: **plus, total** or **sum**
- **Subtraction** may be phrased using the words: **difference** or **take away**

Examiner Tips and Tricks

A good way to check your answer without a calculator is to estimate it.

E.g. If you work out 32 870 ÷ 865 to be 295, check by doing 30 000 ÷ 1 000 in your head which is 30, so your answer is probably wrong (the actual answer is 38).

Worked Example

(a) Find the sum of 3985 and 1273.

Answer:

Notice that the word sum is used but this means add
Quickly estimate the answer

4000 + 1000 = 5000

Write the numbers in two rows and columns aligned

$$\begin{array}{r} 3985 \\ +1273 \\ \hline \end{array}$$

$$\begin{array}{r} 3985 \\ +1273 \\ \hline 8 \end{array}$$

Move on to the tens (next on the left) column
The sum is 15 so the 5 (ones) is written below the line and the 1 (tens) 'carries over' to the next (hundreds) column

$$\begin{array}{r} 1 \\ 3985 \\ +1273 \\ \hline 58 \end{array}$$

Next is the hundreds column which again results in a two-digit answer

$$\begin{array}{r} 11 \\ 3985 \\ +1273 \\ \hline 258 \end{array}$$

Finally add the digits in the thousands column

$$\begin{array}{r} 11 \\ 3985 \\ +1273 \\ \hline 5258 \end{array}$$

Check the final answer is similar to your estimate; 5000 and 5258 are reasonably close

3985 + 1273 = 5258

(b) Find the difference between 506 and 28.

Answer:

Notice that the word difference is used; this means subtract
Quickly estimate the answer

500 - 30 = 470

Write the numbers in two rows, column aligned, ensuring the top number is the number being subtracted *from*

$$\begin{array}{r} 506 \\ -\ 28 \\ \hline \end{array}$$

In the ones (units) column, 6 is smaller than 8, so borrow from the next column (tens)
The tens column is 0, so borrow from the column to the left of that (hundreds)

$$\begin{array}{r} 4 \\ \cancel{5}\,{}^{1}06 \\ -\ 28 \\ \hline \end{array}$$

This turns the 0 (in the tens column) into a 10 which we can then borrow from (for the ones column)

$$\begin{array}{r} 4\ 9 \\ \cancel{5}\,{}^{1}\cancel{0}\ {}^{1}6 \\ -\ 2\ \ 8 \\ \hline \end{array}$$

$$\begin{array}{r} \cancel{5}\,\cancel{1}\,\cancel{0}\,\,{}^{1}6 \\ -\quad 2\quad 8 \\ \hline 8 \end{array}$$

Move onto the tens column which is 9 - 2 = 7
There is nothing to subtract in the last (hundreds) column (4 - 0 = 4)

$$\begin{array}{r} 4\,9\phantom{\,{}^{1}6} \\ \cancel{5}\,\cancel{1}\,\cancel{0}\,\,{}^{1}6 \\ -\quad 2\quad 8 \\ \hline 4\,7\quad 8 \end{array}$$

Check the final answer is similar to the estimate; 470 and 478 are reasonably close

506 - 28 = 478

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